Leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Leakage

> Prepare Leakage: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Choose a split unit that matches the intended **prediction** claim

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/11, ai-4/8. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

Look for: Separate **prediction input**s from observed targets

Reference example, not the only acceptable wording: Candidate features are pages and queue_at_checkout.

The target is returned_late and the tracking field is loan_id. days_to_return is known only after return and directly defines the target, so it must not enter checkout **prediction**s.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

Concept to explain

Leakage occurs when a feature exposes the target through information that would be unavailable, created later, or derived from the answer at real **prediction time**.

Term	Meaning
leakage	target information improperly enters features
prediction time	moment when real input is available
post-outcome feature	field recorded after the target occurs
split leakage	related records cross data splits

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions with fictional frame cards. Focus on the evaluation goal and unit of separation before percentages. The worked plan separates trips for an intended new-trip claim on a familiar route.

It neither trains a model nor establishes performance. Distinguish frames, **group**s and sets, and retain Week 5 label-rule and review records.

Readiness answer

They can share the same mistake or use an unclear rule. Agreement counts matching labels; correctness requires checking the rule and **evidence**.

Printed lesson and workbook

Twelve frame cards A1 A2 A3 B1 B2 C1 D1 D2 E1 E2 F1 F2

Three trays or paper areas named **training** validation and test

Optional Python 3 editor with no extra packages

Apply the connected idea: Scaling without leakage

Teach two 45-minute sessions. Keep the two feature columns distinct and require students to state which rows supplied each fitted number. First compare hand calculations, then trace optional Python.

Explain that min-max scaling is one pre**process**ing choice; a small demonstration cannot establish the best method for a real task.

Readiness answer

The selected features, distance measure and declared tie rule determine the nearest neighbour. Only **training** examples belong in the stored pool; validation reference labels must not enter it.

Printed lesson and workbook

Two coloured number lines for the feature columns

Optional Python 3 editor with no packages required

Blank table for minima **range**s **transform**ed pairs and distances

End goal for the pictorial case

By the end, I can explain **leakage** using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Leakage using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Find and explain four **leakage** paths in a school-event dataset.

MODEL IDEA **Leakage** occurs when a feature exposes the target through information that would be unavailable, created later, or derived from the answer at real **prediction time**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD A leaf file name contains 'diseased'—the answer is hidden in the **input**. Treatment outcome was recorded after diagnosis, so it is unavailable at **prediction time**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 A leaf file name contains 'diseased'—the answer is hidden in the **input**.

Step 2

STEP 2 Treatment outcome was recorded after diagnosis, so it is unavailable at **prediction time**.

Step 3

STEP 3 Near-duplicate photos of one leaf appear in train and test.

Step 4

STEP 4 Remove the leaking fields, **group** related photos before splitting, retrain, and reevaluate.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Ten photos per leaf are randomly split so one leaf appears in train and test.

The teaching case

Explain and repair **leakage**.

Exact explanation to model

Why the result follows

Related views make evaluation familiar. Split by leaf identity before **fitting** or learned pre**process**ing; keep all views together.

Explain the added worked example

Explain the added worked example · 1

A model may seem successful when its test examples closely resemble examples it already learned from. Imagine several pictures taken seconds apart during one robot trip.

They have different picture IDs but share a route, lighting and objects. This week you will assign entire trips to separate sets and explain what the resulting evaluation could measure. No model is fitted in this lesson.

Explain the added worked example · 2

Our invented twelve-row dataset contains frame descriptions from six trips. Each row is one frame; the **group** is its trip ID. The proposed claim concerns new trips on the same practice route under similar conditions.

We therefore keep all frames from a trip in one set. To claim success on new routes, **group**ing only by trip would not be enough.

This small fictional assignment is for learning split design, not **evidence** of real performance. Each frame belongs to one **group** and one set.

Store frame and trip IDs as metadata for auditing; do not assume an ID is an appropriate **prediction input**.

Explain the added worked example · 3

Training data fits the model. Validation data helps compare choices such as candidate rules. Keep the **test set** for a final check after those choices are fixed.

A learner who repeatedly changes the model after seeing test results has used that set for development; its score is no longer an untouched final check.

The first line of **output** is []: no trip ID is shared between any pair of sets. The second is 3 1 2: **training**, validation and test contain three, one and two trip **group**s. The symbol & finds shared values and | combines sets.

These are **group** counts, not frame counts.

This audit assumes the lists correctly describe every frame assignment. It cannot find an omitted frame, a wrong trip ID, a copied image with a different ID or a feature containing the answer.

Check the assignment table and source records too.

Key terms

Term	Meaning
Training set	Examples used to fit a model

Term	Meaning
Validation set	Separate examples used to compare choices during development
Test set	Held out examples used for final evaluation after choices are fixed
Group	Related examples that must stay together for a stated evaluation goal

Apply the connected idea: Scaling without leakage

Last week we compared one numeric feature. With two features, their numerical **range**s can give one feature much more influence. Scaling changes those contributions.

We will learn a minimum and **range** for each feature from **training** only, then reuse them for every example.

Apply the connected idea: Scaling without leakage · 4

These fictional trips have a distance reading in cm, a brightness reading in sensor units and a reviewed lane label. Features are available before **prediction**; labels are separate **evidence**.

Related trips have already been **group**ed before splitting. For this exercise use k=1, add absolute differences across the two features, and break equal distances by alphabetical ID.

Apply the connected idea: Scaling without leakage · 5

For query (11,120), raw distances to A,B,C are 1+80=81, 9+20=29 and 9+80=89. B wins and predicts blocked. Adding these different units is an unscaled numerical choice; the large brightness **range** strongly affects it.

Scaling offers another explicit distance choice, not proof of which features matter.

Apply the connected idea: Scaling without leakage · 6

For each feature use scaled value = (value minus **training** minimum) divided by **training range**. **range**(2) visits feature positions 0 and 1.

The code prints [10, 100] [10, 100], then [(0.0, 1.0), (1.0, 0.0), (1.0, 1.0)], then (0.1, 0.2). Each minimum becomes 0 and each maximum becomes 1.

Apply the connected idea: Scaling without leakage · 7

The first calculations fit low and span from train. **transform** uses those stored values without learning new ones. The code assumes nonempty **training** rows with exactly two finite numeric features; it does not check all **input**s.

Its explicit zero-**range** guard stops division by zero so a constant feature can be reviewed. This is a classroom policy, not a claim that every library stops this way.

Apply the connected idea: Scaling without leakage · 8

Guided: raw distances A,B,C are 81,29,89 so B predicts blocked; scaled distances are 0.9,1.1,1.7 so A predicts clear. Keep minima (10,100), **range**s (10,100), feature order and method settings fixed.

Exit: even unlabelled evaluation features affected the fitted **transform**ation. Split first, fit minima and **range**s on **training**, **transform** every split with those values, then evaluate frozen **prediction**s.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model the reasoning aloud · 9

Suppose A1 and A2 are in **training** while A3 is in test. The frame IDs differ, but A appears in both sets. Shuffling the rows does not remove this relationship.

For our new-trip claim, move all A frames to one chosen role and record the revised assignment before evaluating a model. Do not copy A3 into both roles.

Model the reasoning aloud · 10

Audit that every source frame occurs exactly once, no **group** crosses roles and the assigned counts sum to the source count. In our worked plan the row counts are six **training**, two validation and four test: twelve total.

A fixed random seed can reproduce a random split, but it cannot make a flawed **group**ing choice fair.

A learner moves D2 from validation to training but leaves D1 in validation. Give the new row counts and shared trip ID. Explain why different frame IDs and a fixed seed do not repair the overlap.

Model the reasoning aloud · 11

A split record names the **prediction** goal, source version, **group** definition, exact assignments, row and **group** counts, date and any random method or seed used.

For this hand-assigned example, record that there was no random sampling. Keep label-rule versions and unresolved review statuses traceable as in Week 5.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Leakage

Split by plant identity when testing generalisation to new plants.

The facts we will use

Supplied facts

Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/8

Split by plant, not by view

	view1	view2	view3
Cedar	train	train	train
Elm	train	train	train
Fir	test	test	test

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/8

Leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Group related views

All three Cedar views stay together; likewise Elm and Fir.



2. Assign groups

Training receives six views from Cedar/Elm; test receives three Fir views.



3. State claim

The final test probes one unseen plant, not three independent plants.

CHANGE ONE THING

Move Fir's first view into training while retaining its other two in test.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group related views	All three Cedar views stay together; likewise Elm and Fir.
2. Assign groups	Training receives six views from Cedar/Elm; test receives three Fir views.
3. State claim	The final test probes one unseen plant, not three independent plants.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A correct **group** split still needs enough representative plants; this tiny exercise is not a

reliable population estimate.

Change one thing in the pictured case

Change one thing

Change: Move Fir's first view into **training** while retaining its other two in test.

Predicted result

Prediction: The unseen-plant evaluation is compromised.

Why the result follows

Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Ask while showing the picture

Ask while showing the picture · 12

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Leakage, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Ten photos per leaf are randomly split so one leaf appears in train and test.

2. State the decision to explain

Write the question to explain: Explain and repair **leakage**.

3. Reveal the worked reasoning

Reveal after **prediction**s: Related views make evaluation familiar. Split by leaf identity before **fitting** or learned pre**process**ing; keep all views together.

Hold back the independent task

Keep for an independent attempt: A feature records the expert's final species decision, made after the **prediction time**. Why is this **leakage**, and what should happen to it?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: the row counts become 7 **training**, 1 validation and 4 test. D is shared by **training** and validation because D1 and D2 are related frames from one trip. Unique frame IDs and a seed do not remove the shared trip.

Return D2 to validation to recover the worked plan. Exit: separate trip IDs on one route do not test unseen routes.

Hold out whole routes with all their trips, and collect enough routes and relevant conditions for the intended evaluation.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve why matching labels can share an error. State the new-trip evaluation goal and define one row and one group.
7 to 18 minutes	Use the frame cards to build the worked split. Count rows and groups separately and explain the three roles.
18 to 32 minutes	Trace the intersection audit. Complete W1 and W2; ask which metadata problems an empty overlap list could miss.

Stage	Teacher action
32 to 45 minutes	Move D2 into training and complete the guided question. Repair it by moving it back, then record the before and after counts.
Session 2 start to 8 minutes	Retrieve why shuffling and fixed seeds do not remove relationships. Ask what must stay together for a new-route claim.
8 to 20 minutes	Complete W3 and discuss group-based, route-based and time-respecting evaluation goals. Explain why percentages cannot decide the goal.
20 to 35 minutes	Collect W4 independently before hints. Require all three overlap checks, exact repaired frame lists and both count units.
35 to 45 minutes	Discuss W5 and the exit question. Use the fresh retry after support and record what help was needed.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve k=1 prediction and training-only neighbour storage. Introduce two features with different ranges.
7 to 18 minutes	Calculate raw distances and discuss how units affect contributions. Complete W1 minima and ranges.
18 to 32 minutes	Transform A,B,C and the query. Trace fit versus transform in code and finish W1.
32 to 45 minutes	Complete W2 and the guided question. Use the diagram to follow the same stored scale across splits.
Session 2 start to 8 minutes	Retrieve the stored feature order, minima and ranges without recalculating from validation.
8 to 20 minutes	Transform and score all three validation rows. Complete W3 and explain values outside 0 to 1.
20 to 35 minutes	Collect W4 independently before giving support. Require raw and scaled arithmetic plus the new row.
35 to 45 minutes	Discuss W5 and the exit question. Use the fresh retry after feedback and record support.

Reduce help deliberately

Completion practice

Completion task: For the worked A through F plan, give **training**, validation and test counts first in **group**s and then in rows. Interpret [] from the code and state one problem it cannot detect.

Reasoning to listen for

Teacher check: **Group** counts are 3,1,2; row counts are 6,2,4. The empty list means no trip ID overlaps between any pair of the listed sets.

The code cannot detect wrong metadata, an omitted frame, a disguised duplicate or an outcome feature.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Different frames may share a trip or source. Choose **group** metadata based on the claim and verify it against the source.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted **group** explanation while others solve a more demanding variation.

Connected teaching cycle: Scaling without leakage

Target: Calculate feature scales from **training** data only

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: For query (11,120), calculate raw and scaled distances to A,B,C. Give both k=1 **prediction**s. Does this prove that scaling always improves **prediction**s?

Reasoning to listen for

Teacher check: Raw distances 81,29,89 select B blocked. Scaled distances 0.9,1.1,1.7 select A clear. The feature contributions changed.

This single changed **prediction** does not establish better performance; evaluate separately and report limits.

Individual check: For Scaling without **leakage**, use its worked case to evaluate this proposed explanation: “Every **transform**ed value must lie between 0 and 1”. Explain the deciding step.

Expected repair

Expected correction: Only the fitted **training range** maps to 0 through 1. New values outside that **range** can **transform** below 0 or above 1.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

One: Fir. Its three photos are related views, so report both photo count 3 and plant count 1. Changed-case check: The unseen-plant evaluation is compromised.

Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Changed-case reasoning: The unseen-plant evaluation is compromised. Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Leakage investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. **Leakage** Detective: draw a timeline for every suspicious field, inspect **group** identity, and repair the split before **training**.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A highly accurate feature should be kept even when it would not exist during real use.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Different row IDs mean independent examples: Different frames may share a trip or source. Choose **group** metadata based on the claim and verify it against the source.

A fixed percentage guarantees a fair test: Percentages do not remove overlap or ensure enough **group**s and relevant cases. The goal and dependencies determine the split.

An empty overlap list proves there is no **leakage**: The audit checks listed **group** IDs only. Wrong metadata, copied content, outcome features or pre**process**ing fitted on all sets can still invalidate evaluation.

Check these misconceptions

Every **transform**ed value must lie between 0 and 1: Only the fitted **training range** maps to 0 through 1. New values outside that **range** can **transform** below 0 or above 1.

No labels means no **leakage**: Held-out feature values can leak into a learned **transform**ation. Fit pre**process**ing on the **training** split only.

Scale each split separately: Separate fits create different coordinates. Apply the same stored scale to **training** examples and queries.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Leakage questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Related views make evaluation familiar. Split by leaf identity before **fitting** or learned pre**process**ing; keep all views together.

Independent question

Independent question · withhold the key

A feature records the expert's final species decision, made after the **prediction time**. Why is this **leakage**, and what should happen to it?

Expected independent answer

Teacher key · Expected independent answer

It exposes the answer or information unavailable at **prediction time**. Exclude it and

reevaluate using only legitimately available features.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Any recorded column is a valid predictor regardless of when it becomes available.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Views of the same leaf can appear in both sets, making evaluation familiar. Assign all views of each leaf to one partition before **fitting** or augmentation.

If deployment concerns new plants or sites, consider **group**ing at that broader level too.

Explanations for the additional workbook examples

W1 Explain the three roles

Workbook question · W1

Explain what **training**, validation and **test set**s are used for. What changes if you repeatedly choose a model using its test score?

Teacher reasoning · W1

Training fits the model; validation compares development choices; test evaluates after choices are fixed. Repeatedly selecting by the test score uses that set for development, so it is no longer an untouched final evaluation.

W2 Distinguish frames from groups

Workbook question · W2

For the worked A through F plan, give **training**, validation and test counts first in **group**s and then in rows. Interpret [] from the code and state one problem it cannot detect.

Teacher reasoning · W2

Group counts are 3,1,2; row counts are 6,2,4. The empty list means no trip ID overlaps between any pair of the listed sets. The code cannot detect wrong metadata, an omitted frame, a disguised duplicate or an outcome feature.

W3 Repair a related example leak

Workbook question · W3

A1 and A2 are in **training** and A3 is in test. Explain why unique frame IDs and a fixed random

seed do not fix this. Propose a repair and what to record.

Teacher reasoning · W3

The frames share trip A, so the test contains a trip already represented in **training**. Unique frame IDs do not remove similarity; a seed only reproduces the assignment.

Put every A frame in one role, keep each frame once and record the revised assignments and counts before model evaluation.

W4 Audit and repair independently

Workbook question · W4

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

Teacher reasoning · W4

Training/validation share P; **training**/test share P and Q; validation/test share P and S. Repaired **training** is P1 P2 P3 Q1 Q2 (5 rows,2 **group**s), validation R1 (1 row,1 **group**), test S1 S2 (2 rows,1 **group**).

Every frame appears once and **group**s are disjoint. This tiny plan does not establish reliable performance, correct metadata, label coverage or success on new routes.

W5 Match separation to the claim

Workbook question · W5

A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative.

Is this enough for a strong performance claim?

Teacher reasoning · W5

Both routes occur in **training** and test, so the plan does not hold out a new route. For illustration, keep A and B together for **training** and C and D together for test, or reverse the route roles.

With only two routes there is too little **evidence** for a strong general claim and no separate route left for validation; obtain more routes for a fuller design.

Scaling without leakage — W1 Learn the scale from training

Using A,B,C, give the minimum, maximum and **range** for each feature. **Transform** all three **training** rows and query (11,120). Explain which rows supplied the scale.

Distance minimum 10 maximum 20 **range** 10; brightness minimum 100 maximum 200 **range** 100. A becomes (0,1), B (1,0), C (1,1), and query (0.1,0.2). Only A,B,C supplied the scale; the query did not.

Scaling without leakage — W2 Trace the changed prediction

For query (11,120), calculate raw and scaled distances to A,B,C. Give both k=1 **prediction**s. Does this prove that scaling always improves **prediction**s?

Raw distances 81,29,89 select B blocked. Scaled distances 0.9,1.1,1.7 select A clear. The feature contributions changed.

This single changed **prediction** does not establish better performance; evaluate separately and report limits.

Scaling without leakage — W3 Keep one coordinate system

Transform (12,80) using the stored **training** scale. Explain the negative value and why validation must not fit its own scale. Give the worked validation and baseline scores.

(12,80) becomes (0.2,-0.2). Brightness is below the **training** minimum, so the negative result is valid under this formula. Re**fitting** validation creates different coordinates; use the stored **training** **transform**ation.

Scaled k=1 scores 2/3 and the blocked baseline 1/3.

Scaling without leakage — W4 Scale a fresh dataset independently

New **training** rows are P (20,1000) clear, Q (40,600) blocked, R (40,1000) blocked. Fit minima and **range**s. For query (22,680), show **transform**ed features, raw and scaled distances to P,Q,R, and both k=1 **prediction**s.

Transform new row (50,520) with the same scale.

Minima (20,600), **range**s (20,400). P,Q,R become (0,1),(1,0),(1,1); query becomes (0.1,0.2). Raw distances 322,98,338 select Q blocked. Scaled distances 0.9,1.1,1.7 select P clear.

New row (50,520) becomes (1.5,-0.2); keep the fitted scale and do not silently clip.

Scaling without leakage — W5 Repair two preprocessing mistakes

A learner fits scales using **training** plus validation features but no labels. Another divides by a zero **training range**. Explain both problems and give a recorded repair for each.

Evaluation features influenced the first learned **transform**ation, causing **leakage** even without labels. Split first, fit only on **training** and reuse those values for validation. A zero **range** causes division by zero.

This code stops for review; one documented repair is to omit that feature consistently from every split and update the feature list.

Assess independent transfer

Independent question · withhold the key

Selected case: Each of five plants has four photos. A random photo split puts different views of every plant in both **training** and **test set**s. Explain the problem and repair.

Selected case answer

Teacher key · Selected case answer

Related views of the same plant cross the split, making the test less independent. Split by plant identity and keep all four views together.

Unique filenames do not make plants independent; declare plant assignments before **fitting** or tuning.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Leakage to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed **leakage** path.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and sub**group** or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 9: Dataset Cards. Retrieve today’s **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Keep a coloured border shared by cards from the same trip. Supply blank assignment tables without the correct roles.

Extension: explain why a new-route evaluation with many frames but only two routes has limited **evidence** and cannot obtain three nonempty disjoint route sets.

Require W4 overlaps P; P and Q; P and S, in pair order, then exact repaired lists with rows 5,1,2 and **group**s 2,1,1. Fresh retry: U1 U2 belong to U, V1 to V, W1 W2 to W.

Proposed **training** U1 V1, validation W1, test U2 W2 has shared U between **training**/test and W between validation/test; **training**/validation share none.

Repair to all U in **training**, V in validation and all W in test: rows 2,1,2 and **group**s 1,1,1. Require an explicit small-sample limitation.

Week 7 introduces a baseline before a model. Carry forward the split record and keep the final test protected while comparing choices.

Teacher reference: scikit-learn cross-validation documentation, **group**ed data section, https://scikit-learn.org/stable/modules/cross_validation.html (reviewed 2026-09-27).

Readiness and reassessment for Scaling without leakage

Colour each feature column separately and supply the formula with spaces for the **training** minimum and **range**. Let students use fractions before decimals.

Extension: convert one entire feature to a different positive unit, refit on **training** only, and explain why the scaled values remain the same.

Require W4 minima (20,600), **range**s (20,400), query (0.1,0.2), raw distances 322,98,338 and scaled distances 0.9,1.1,1.7. **Prediction**s Q blocked and P clear; new row (1.5,-0.2).

Fresh retry: L (5,80) blocked, M (15,40) clear, N (15,80) clear; query (6,48). Minima (5,40), **range**s (10,40), query (0.1,0.2). Raw distances 33,17,41 select M clear; scaled distances 0.9,1.1,1.7 select L blocked.

Require **training**-only **fitting** and consistent **transform**ation.

Plan the delayed return

Suggested return points in this level: ai-4/9, ai-4/11, ai-4/14. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Leakage lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Choose a split unit that matches the intended **prediction** claim

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: For the worked A through F plan, give **training**, validation and test counts first in **group**s and then in rows. Interpret [] from the code and state one problem it cannot detect.

Supported practice answer

Reasoned practice answer: **Group** counts are 3,1,2; row counts are 6,2,4. The empty list means no trip ID overlaps between any pair of the listed sets.

The code cannot detect wrong metadata, an omitted frame, a disguised duplicate or an outcome feature.

Check every learner before discussion: Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Different frames may share a trip or source. Choose **group** metadata based on the claim and verify it against the source.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Move Fir's first view into **training** while retaining its other two in test. Require a **prediction** and a reason before comparing. Expected result: The unseen-plant evaluation is compromised.

Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

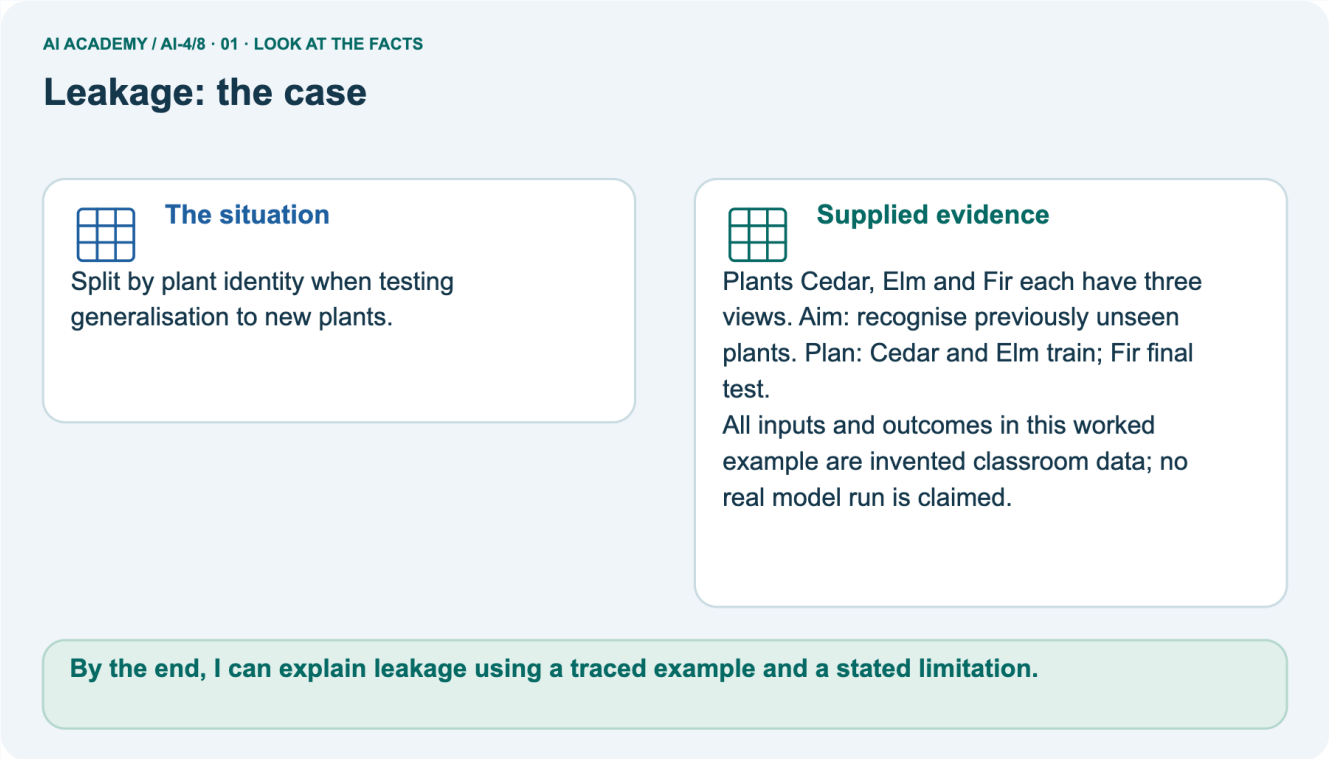
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Leakage graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Leakage: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant identity when testing generalisation to new plants. Supplied facts: Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

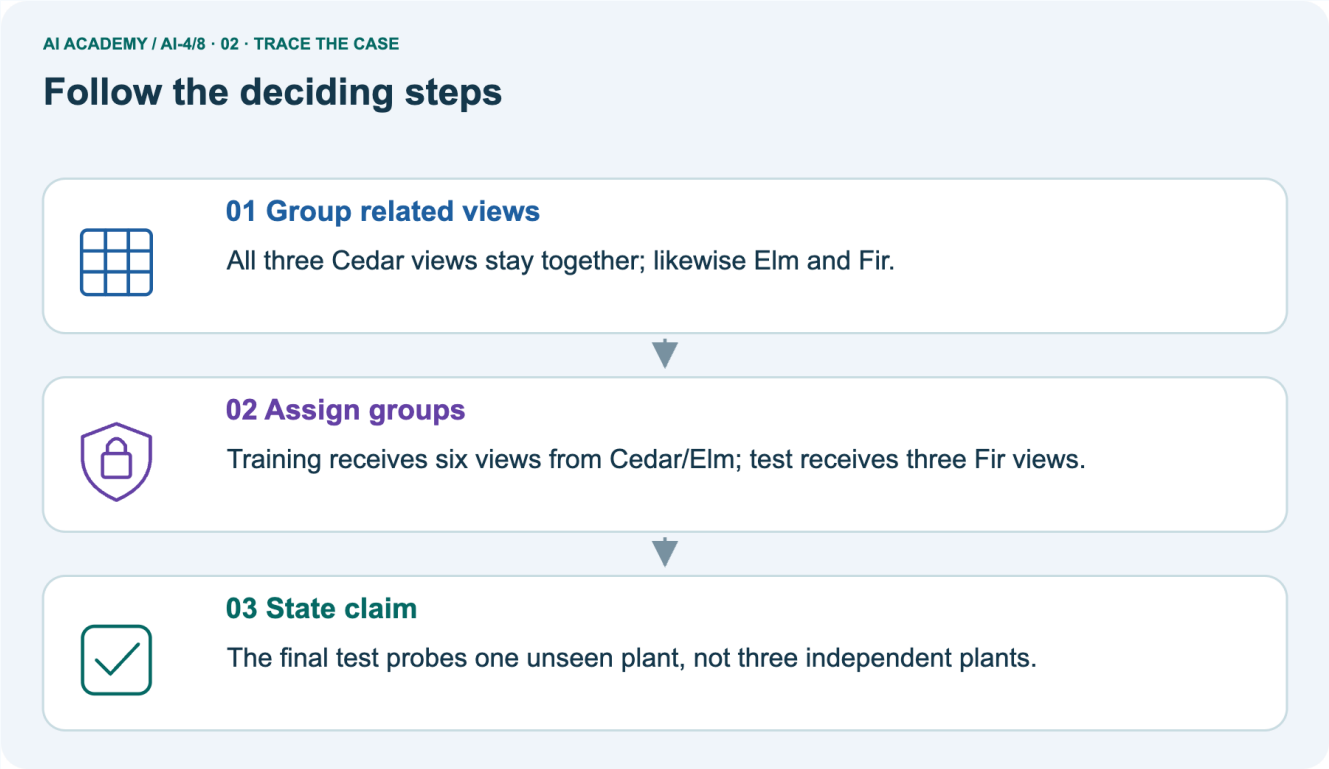
> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Group related views: All three Cedar views stay together; likewise Elm and Fir. → Assign **group**s: **Training** receives six views from Cedar/Elm; test receives three Fir views. → State claim: The final test probes one unseen plant, not three independent plants.

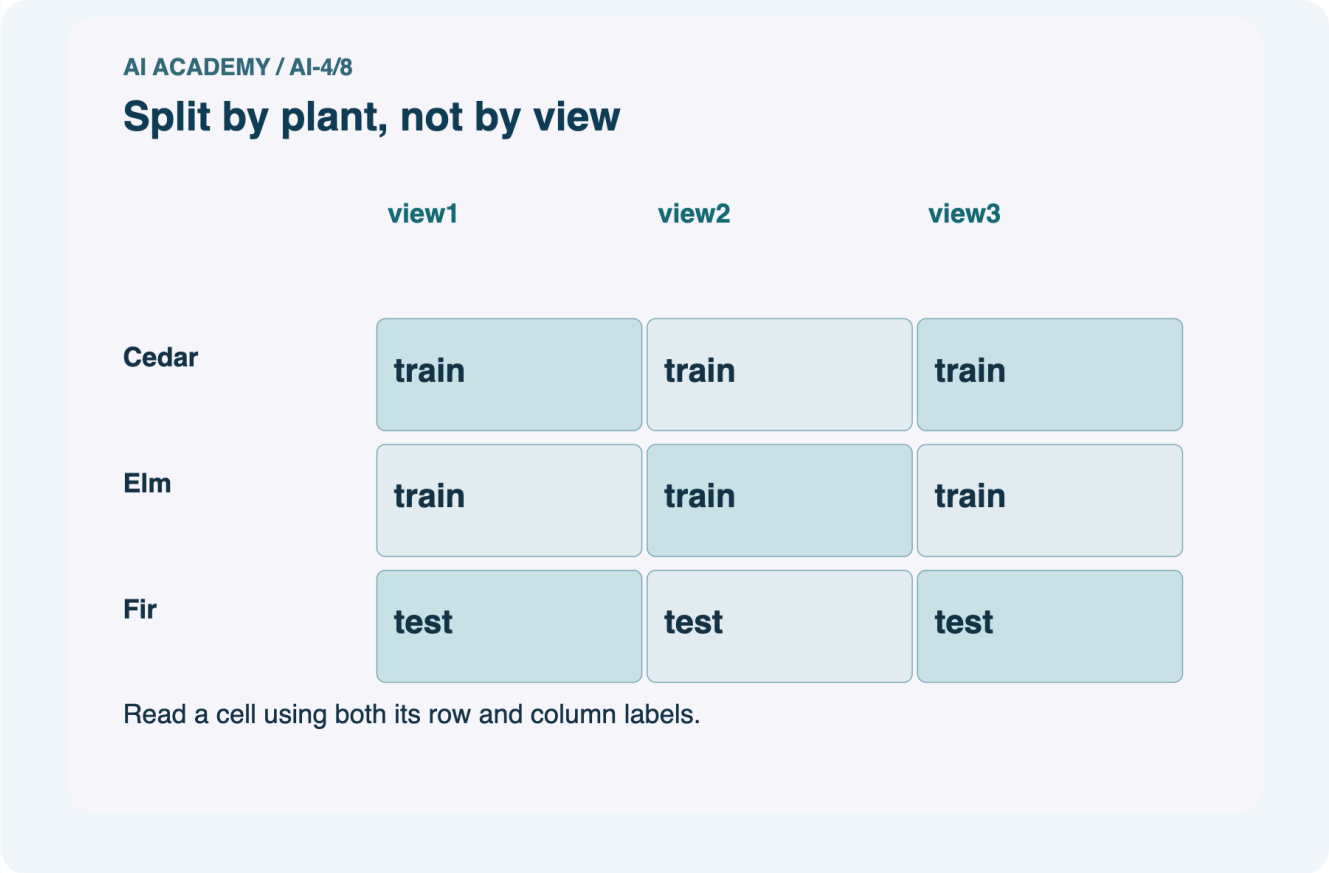
Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if

the image is difficult to see.

03. Split by plant, not by view



Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant, not by view. Cedar: view1 = train, view2 = train, view3 = train. Elm: view1 = train, view2 = train, view3 = train. Fir: view1 = test, view2 = test, view3 = test.

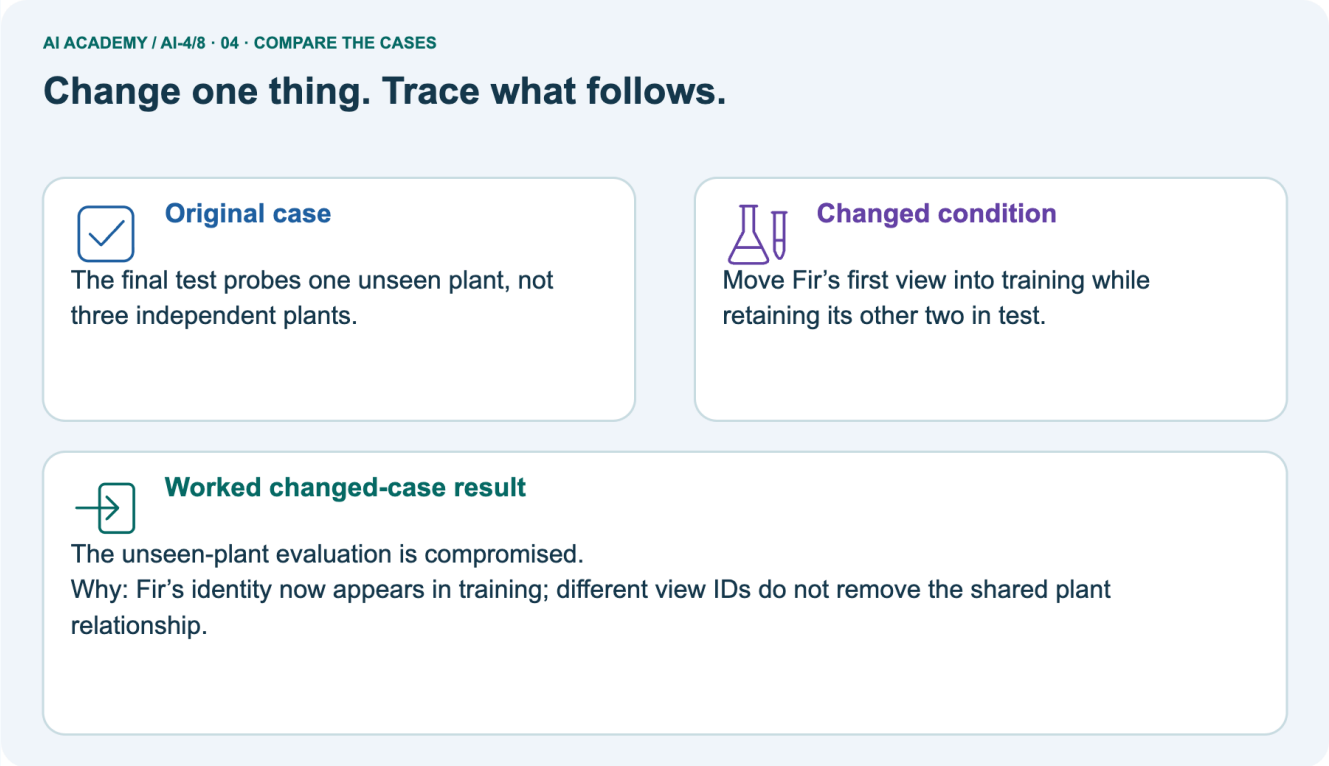
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The final test probes one unseen plant, not three independent plants. Changed condition: Move Fir's first view into **training** while retaining its other two in test.

Expected result: The unseen-plant evaluation is compromised. Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

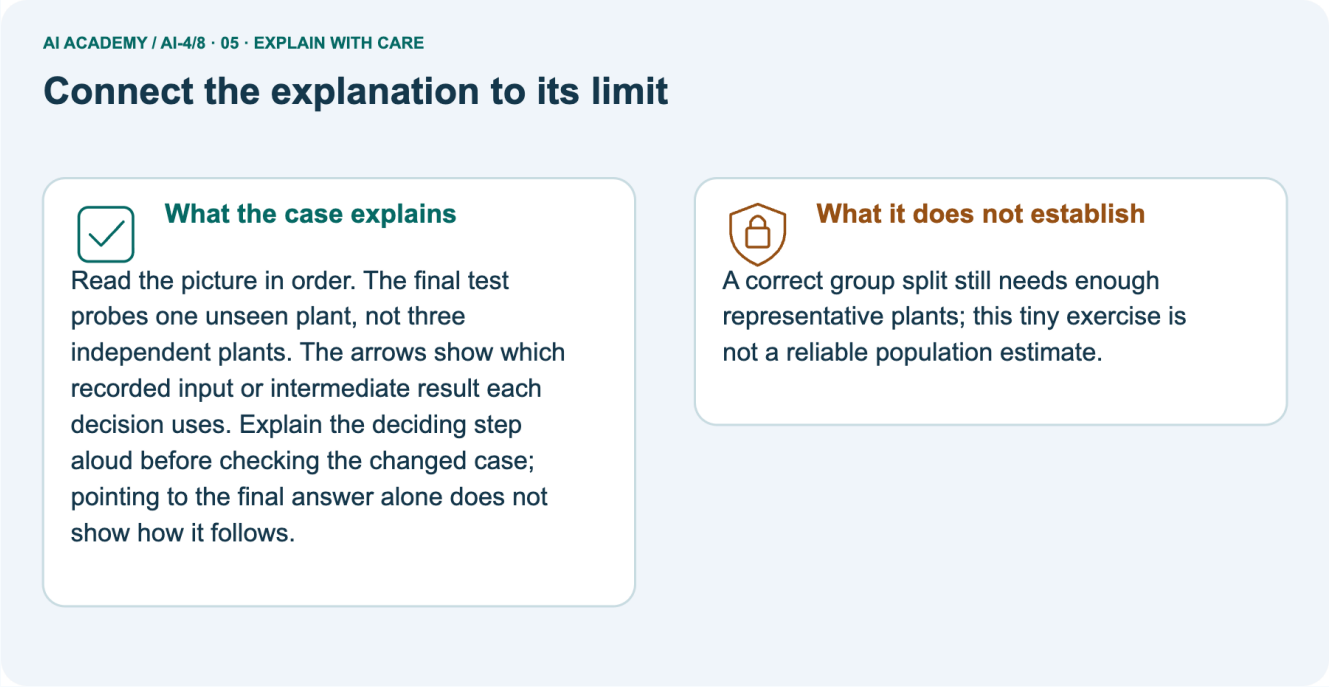
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correct **group** split still needs enough representative plants; this tiny exercise is not a reliable population estimate.

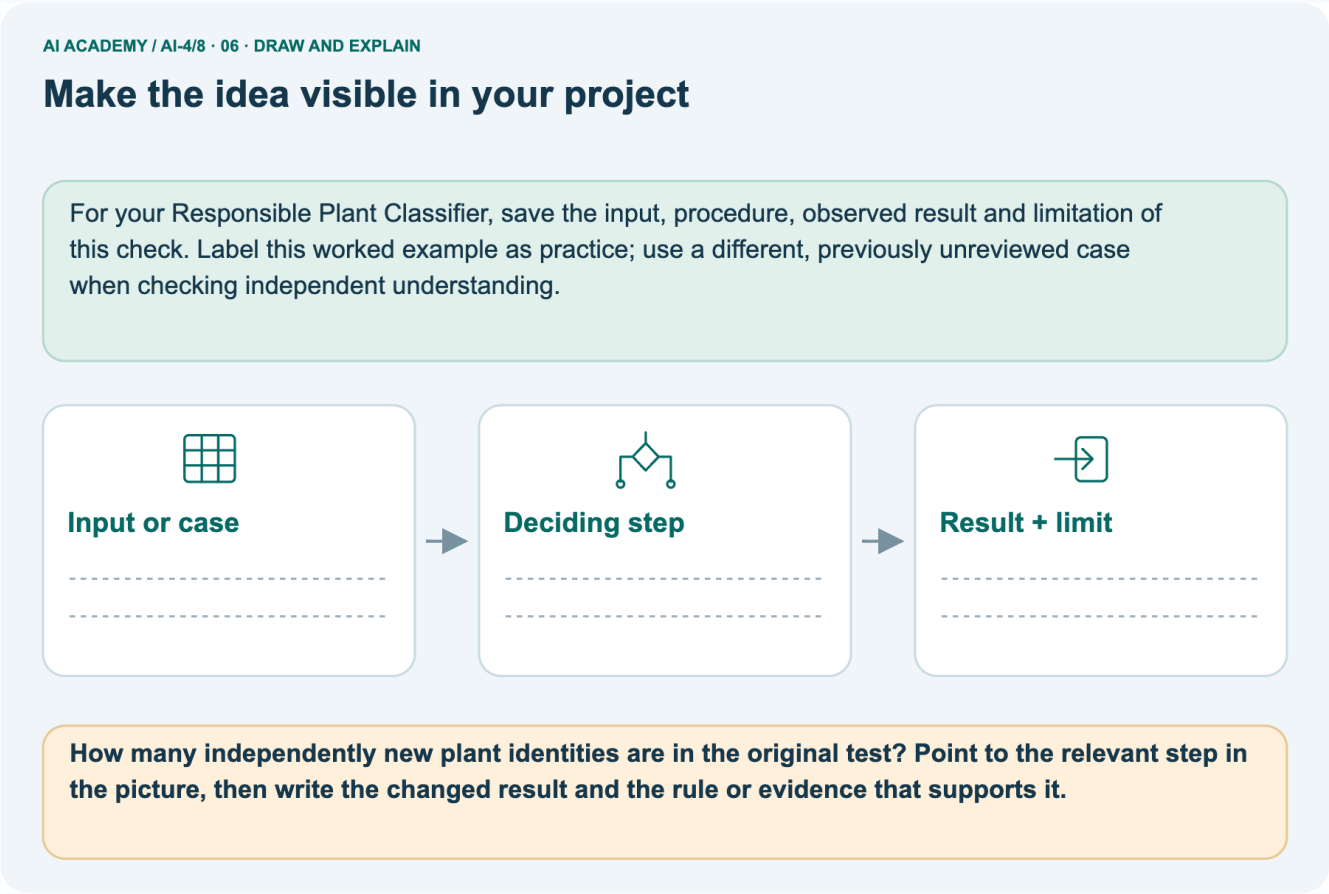
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A correct **group** split still needs enough representative plants; this tiny exercise is not a reliable population estimate.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: One: Fir. Its three photos are related views, so report both photo count 3 and plant count 1. Changed-case check: The unseen-plant evaluation is compromised.

Fir’s identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.